

Internship in Physical Organic Chemistry

Week 2 — Part 3: Spectral Acquisition and Structure Characterisation

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Goals of this part:

- Record a complete set of MS1 and MS/MS spectra for your compound
- Identify and assign the major ions in your spectra
- Confirm the molecular formula using isotope simulation
- Identify the retro-Diels-Alder fragment ions and their neutral losses
- Cross-compare positive and negative ion mode assignments
- Complete scratch sections 2W-b, 2W-c, and 2W-d

1 Acquisition workflow

Before sitting down at the instrument, confirm that you have completed the pre-acquisition checklist from Part 2: solvent and concentration confirmed with supervisor, sample prepared and labelled, lab journal template printed, folder structure created, file naming convention understood.

1.1 Order of acquisition

Acquire spectra in the following order. Record every spectrum in your lab journal as you go, including the file name, ionisation mode, and any observations.

1. **ESI optimisation — positive ion mode.** Before acquiring any spectra for analysis, optimise the source conditions for your compound in positive ion mode. Set up a continuous infusion of your sample and monitor the MS1 spectrum in real time. Vary the **spray voltage** across 2–3 values within the range indicated by the engineer during the training day. For each value, note the intensity of the molecular ion ($[M+H]^+$) in your lab journal. Then vary the **in-source voltage** (skimmer offset or equivalent parameter — the exact name is confirmed during the training day) across 2–3 values. Observe what happens to the MS1 spectrum: at low values the spectrum should show primarily the molecular ion and any adducts; at higher values, fragment ions will begin to appear in the MS1 spectrum. This is called **in-source fragmentation** and must be avoided during acquisition — it produces ions that look like MS/MS fragments but are not controlled and cannot be used for energy-resolved measurements. Choose the in-source voltage value that gives a clean MS1 with minimal in-source fragmentation. Fix the spray voltage and in-source voltage that give the best signal for the molecular ion without in-source fragmentation. Record the chosen values in your lab journal and in scratch section 2W-exp.
2. **Positive ion MS1 (full scan).** At the optimised source conditions, acquire a full-scan MS1 spectrum over the full mass range (typically m/z 100–1000). Save the raw file immediately using the naming convention from Part 1.

3. **MS/MS of every peak observed in positive MS1.** Identify every peak with a signal above the noise baseline in the positive MS1 spectrum. For each peak, select it as the precursor ion and acquire an HCD MS/MS spectrum. Use three NCE values: a low value (NCE 10–15), a medium value (NCE 25–30), and a higher value (NCE 40–50). This range will show you how fragmentation changes with energy — at low NCE you may see little fragmentation; at higher NCE competing pathways open up. Save a separate raw file for each precursor and each NCE value, following the naming convention.
4. **ESI optimisation — negative ion mode.** Switch to negative ion mode. Repeat the optimisation procedure from step 1: vary the spray voltage and in-source voltage, and fix the conditions that give the best signal for $[M-H]^-$ without in-source fragmentation. Note that the optimal spray voltage in negative mode will differ from positive mode. Record the chosen values in your lab journal and in scratch section 2W-exp.
5. **Negative ion MS1 (full scan).** At the optimised negative-mode source conditions, acquire a full-scan MS1 spectrum. Save the raw file.
6. **MS/MS of every peak observed in negative MS1.** Repeat step 3 for all peaks observed in the negative MS1 spectrum, using the same three NCE values.

Why in-source fragmentation matters for this project:

The energy-resolved measurements in Week 3 rely on a clean, well-defined precursor ion. If in-source fragmentation is present, the ion entering the HCD cell is already partially degraded, and the breakdown curve will not reflect the true activation barrier. Always verify that your MS1 spectrum is clean before recording MS/MS spectra. A clean MS1 shows the molecular ion as the dominant peak, with adducts and isotope peaks only — no unexplained low-mass ions.

Record every raw file name in your lab journal as you save it. If you acquire a spectrum and decide it is not useful, keep the file anyway — do not delete raw data. Write a note in the lab journal explaining why the spectrum was not used. A missing file is harder to explain than a poor spectrum.

1.2 What “done” looks like

Before leaving the instrument, verify that your **week02/raw/** folder contains:

- ESI optimisation files for positive mode (spray voltage and in-source voltage series)
- One positive MS1 file at optimised conditions
- At least three MS/MS files per peak observed in positive MS1 (one per NCE value)
- ESI optimisation files for negative mode
- One negative MS1 file at optimised conditions
- At least three MS/MS files per peak observed in negative MS1 (one per NCE value)

Export all raw files to Excel immediately after acquisition while still at the instrument, following the procedure from Part 1. Place the exported files in **week02/exported/**.

Fill in scratch sections 2W-b and 2W-c (MS1 observations in positive and negative mode) before leaving the instrument. Write one sentence per observed peak describing what is observed — m/z value, approximate relative intensity, and whether the peak is the dominant ion or a minor species. No interpretation at this stage.

2 Reading the spectra

Before moving to systematic structure characterisation, you must be able to read what Xcalibur is showing you. This section explains the key features to identify in your MS1 and MS/MS spectra.

2.1 The MS1 spectrum: what to look for

In a positive-ion ESI-MS1 spectrum of a small organic molecule, you expect to see:

- **The protonated molecule** $[M+H]^+$: the most common ion in positive ESI. Its m/z equals the molecular mass plus one proton mass (1.0073 Da). This is usually the dominant peak for a pure compound at the concentrations used here.
- **Sodium adduct** $[M+Na]^+$: common in positive ESI, appears 21.982 Da above $[M+H]^+$. Its presence confirms the molecular mass independently.
- **Potassium adduct** $[M+K]^+$: less common, appears 37.956 Da above $[M+H]^+$.
- **Solvent clusters and background ions**: peaks at low m/z (below 150) that do not shift when you change the compound. These are instrument background and should not be selected as precursors for MS/MS.

In negative-ion ESI-MS1, the most common ion is the deprotonated molecule $[M-H]^-$, at m/z equal to the molecular mass minus one proton mass.

Expected ions for this project:

Luteolin has the molecular formula $C_{15}H_{10}O_6$, molecular mass 286.048 Da.

Quercetin has the molecular formula $C_{15}H_{10}O_7$, molecular mass 302.043 Da.

In positive mode, the dominant ion is typically $[M+H]^+$.

In negative mode, the dominant ion is typically $[M-H]^-$.

Use these values to identify your molecular ion before proceeding to MS/MS. If the dominant peak does not match either $[M+H]^+$ or $[M+Na]^+$, consult your supervisor before continuing.

2.2 Confirming the molecular formula by isotope simulation

The Orbitrap measures m/z values with sufficient accuracy (within a few ppm) to confirm the molecular formula of an observed ion. Xcalibur provides an isotope simulation tool for this purpose (Figure 1).

To use it, open the isotope simulation panel (accessible via the MS toolbar button in Qual Browser). Enter the molecular formula of your compound and select the appropriate adduct ($[M+H]^+$ for positive mode, $[M-H]^-$ for negative mode). The tool will display the expected isotope pattern: the monoisotopic peak and the ^{13}C isotope peak (M+1), which for a molecule of 15 carbons will appear at approximately 16% relative intensity.

Compare the simulated m/z value to the observed peak in your spectrum. A match within 5 ppm confirms the molecular formula assignment. Record the observed m/z , the theoretical m/z , and the mass error in ppm in your lab journal.

2.3 The MS/MS spectrum: what to look for

An MS/MS spectrum shows the fragment ions produced when a selected precursor ion is activated by HCD. The precursor ion itself may still be visible at the high- m/z end of the spectrum (as a surviving fraction), with all fragment ions appearing at lower m/z .

The key concept is the **neutral loss**: the difference in m/z between the precursor and a fragment ion. A neutral loss corresponds to a specific chemical group or small molecule that is

Isotope simulation

New Insert Replace

Chemical formula ☒

Peptide/Protein ☐

Plus H₂O ☒ amu

Change mixture...

☐ Adduct

Identity

Concentration

Charge distribution

Most abundant :

Half width:

Output style

☒ Pattern

☐ Profile

Samples/peak:

☐ Centroid

Resolution

☒ 1 Dalton

☐ 50 PPM

☐ 20000 Resolving pwr

Valley

☒ FWHM

☐ 10%

☐ 5%

Help

Figure 1: The isotope simulation panel in Xcalibur. Enter the molecular formula and select the appropriate adduct to simulate the expected isotope pattern. Compare the simulated pattern to your observed spectrum to confirm the molecular formula assignment.

ejected from the precursor. For example, a neutral loss of 28 Da corresponds to loss of CO; a neutral loss of 18 Da corresponds to loss of H₂O.

When you look at an MS/MS spectrum, ask:

1. What is the m/z of the base peak (most abundant fragment)?
2. What neutral loss does each major fragment correspond to?
3. Does the fragmentation pattern change between NCE 10 and NCE 40? Which fragments appear first (low barrier) and which only at higher NCE?

3 Structure characterisation

Once you have acquired and exported all spectra, work through the structure characterisation systematically using the provided templates.

Two standalone templates are provided:

- `structure_char_pos.pdf` — for positive ion mode data
- `structure_char_neg.pdf` — for negative ion mode data

Fill in the positive mode sheet first, then the negative mode sheet. The negative mode sheet includes a cross-comparison section at the end.

3.1 How to use the structure characterisation sheet

The sheet has two parts. The first part is a systematic fragment table: list every observed fragment ion, calculate the neutral loss from the precursor, and propose a chemical assignment. The second part is open space for the RDA argument and structure sketches.

Work through the fragment table methodically:

1. List all fragment ions from the MS/MS spectrum of the molecular ion (or dominant adduct) at the medium NCE value.
2. For each fragment, calculate the neutral loss: $\Delta m/z = m/z_{\text{precursor}} - m/z_{\text{fragment}}$.
3. Look up or calculate the formula corresponding to each neutral loss. Common neutral losses for flavonoids include CO (28 Da), CO₂ (44 Da), and H₂O (18 Da), as well as characteristic ring-opening losses.
4. Mark any fragment you tentatively assign to an RDA pathway.

At this stage you are building a list of observations and tentative assignments. You do not need to be certain about every assignment. Write what you observe and what seems most consistent — you will refine these assignments when you compare with the anchor papers. Distinguish clearly between what you observe (a peak at m/z X) and what you propose (this corresponds to loss of Y).

3.2 Identifying the RDA fragments

The retro-Diels-Alder fragmentation of a flavonoid produces characteristic fragment ions by cleavage of the C-ring. The two resulting fragments retain parts of the A-ring and B-ring respectively, and their m/z values depend on the substitution pattern of the compound.

To identify the RDA fragments in your spectrum:

1. Identify the most abundant fragment ion in the MS/MS spectrum.
2. Check whether it is consistent with a retro-[4+2] cycloreversion of the C-ring: the two fragment masses must sum to the precursor mass (accounting for any proton transfer between the two fragments).

3. Confirm the assignment by comparing with the literature: your Anchor 2 paper reports RDA assignments for the same compound class.
4. Sketch the proposed fragmentation in the open section of the sheet, labelling the A-ring fragment and the B-ring fragment.

A note on the two RDA fragments:

In the RDA fragmentation of flavonoids, one fragment retains the charge and is observed in the spectrum; the other is lost as a neutral. In positive and negative ion mode, different fragments tend to carry the charge. This is one reason why measuring in both modes is informative — the two modes reveal complementary fragments of the same reaction.

3.3 Cross-comparison of positive and negative mode

After completing both structure characterisation sheets, answer the following questions in scratch section 2W-e:

1. Which fragment ions appear in both modes? Which appear in only one mode?
2. Does the RDA cleavage produce the same neutral loss in both modes, or different losses? What does this tell you about which fragment retains the charge in each mode?
3. Is the fragmentation pattern at medium NCE in positive mode richer or poorer than in negative mode? What might explain this?

Complete scratch section 2W-d (MS/MS precursor selection rationale) now: for each precursor ion you selected for MS/MS, write one sentence explaining why you chose it. Then complete section 2W-e (RDA assignment summary): write 2–3 sentences summarising which fragment ions you assign to the RDA pathway and what neutral loss each corresponds to.